Intelligent Network Topology Optimization Using LAM

Karan Khotare (BT22CSE111) Yash (BT22CSE110) Aman Raj (BT22CSE115) Om Ashok Nalage (BT22CSE117)

Department of Computer Science and Engineering

December 6, 2025

Under the guidance of
Dr. Nidhi Lal

Outline

Introduction to ICN Architecture

- Shift from Host-Centric (IP) to Content-Centric (focus on content, not location).

Introduction to ICN Architecture

- Shift from Host-Centric (IP) to Content-Centric (focus on content, not location).
- Routers act as forwarders and in-network caches (dual role).

Introduction to ICN Architecture

- Shift from Host-Centric (IP) to Content-Centric (focus on content, not location).
- Routers act as forwarders and in-network caches (dual role).
- **Packet Types:** Interest (request) and Data (respons with content).

Introduction to ICN Architecture

- Shift from Host-Centric (IP) to Content-Centric (focus on content, not location).
- Routers act as forwarders and in-network caches (dual role).
- **Packet Types:** Interest (request) and Data (respons with content).
- **Router Data Structures:**
    - Content Store (CS)
    - Pending Interest Table (PIT)
    - Forwarding Info Base (FIB)

Introduction to ICN Architecture

- Shift from Host-Centric (IP) to Content-Centric (focus on content, not location).
- Routers act as forwarders and in-network caches (dual role).
- **Packet Types:** Interest (request) and Data (respons with content).
- **Router Data Structures:**
  - Content Store (CS)
  - Pending Interest Table (PIT)
  - Forwarding Info Base (FIB)
- **Communication Flow:** Client sends an Interest, routers forward/cache, the server/cache responds with Data, and the Data packet returns along the path, being cached at intermediate routers.

Problem Statement

Key Challenges

- Traditional networks rely on static caching and routing, leading to inefficient resource utilization.
- Redundancy and high cache churn are common due to inflexible caching policies.

The Need

An intelligent topology optimization that adapts dynamically to network demand and topology changes is required.

Background Concepts

- **Content-Centric Networking (CCN):** Focuses on content availability and retrieval, not the host's location.
- **Centrality Measures:** Metrics to identify the most important or "central" routers in a network topology.
- **CMBA (Centrality-Measures Based Algorithm):** Uses centrality measures to select the most appropriate routers for caching content to improve performance.

Centrality Measure Formulas

Here are the core formulas used to identify influential nodes in the network.

Degree Centrality $C_D(v)$

The number of direct connections a node has.

$$DC(CR_i) = \frac{degree(CR_i)}{(n-1)} \quad (1)$$

Closeness Centrality $C_C(v)$

The reciprocal of the sum of the shortest path distances to all other nodes.

$$CC(CR_i) = \frac{(n-1)}{\sum_{j=1}^{n} d(CR_i, CR_j)} \quad (2)$$

Betweenness Centrality $C_B(v)$

The fraction of all-pairs shortest paths that pass through a node.

$$BC(CR_i) = \sum_{i \neq CR_i \neq j \in n} \frac{\sigma_{i,j}(CR_i)}{\sigma_{i,j}} \quad (3)$$

Reach Centrality $C_R(v)$

It defines how many no. of hops, a particular CR reaches to another CR

$$RC(CR_i) = 1 + \sum_{x=1}^{e(CR_i)} \frac{r_{ix}}{x} \quad (4)$$

Algorithm: Content Request

Algorithm 1: Content request

Initialize (CMV=0)
for for each (CR_n from i to j do) **do**
    if if content item in the CS of CR then **then**
        send towards the client.
        send(data)
    elseelse
        Calculate CMV at each CR upon arriving of interest with respect to a client.
        if if calculated value of CMV $>$ CMV stored in an interest **then**
            Then, update CMV with router-id into interest.
        elseelse
            Forward request to the next hop towards j.
        end if
    end if
end for

Project Idea & Objectives

Core Idea

To develop a Deep Reinforcement Learning (DRL) agent that dynamically optimizes caching and routing decisions in the network.

Objectives

- Implement and evaluate the baseline CMBA.
- Integrate a DRL agent to enhance decision-making.
- Evaluate overall performance improvement in terms of:
    - Cache Hit Ratio (CHR)
    - Hop Reduction Ratio (HRR)
    - Latency & Throughput

Comparison of Caching Strategies

This table shows the performance metrics with and without the Centrality-Measures Based Algorithm (CMBA).

Metric	**Without CMBA**	**With CMBA**
CHR (Cache Hit Ratio)	2.391	2.706
HRR (Hop Reduction Ratio)	0.745	0.843
Latency (ms)	0.1086	0.0943

Table: Performance comparison based on simulation results.

Observation

The introduction of CMBA shows a notable improvement in Cache Hit Ratio and Hop Reduction Ratio, along with a decrease in latency.

Performance Graph: Cache Hit Ratio (CHR)

This graph shows the improvement in Cache Hit Ratio over simulation iterations.

Performance Graph: Hop Reduction Ratio (HRR)

This graph illustrates the reduction in the average number of hops required to fetch content.

Performance Graph: Latency

This graph shows the trend of content retrieval latency across simulation iterations.

Expected Outcomes

- A more intelligent and adaptive network topology.
- Optimized cache placement for better resource utilization.
- Reduced content retrieval latency and redundancy in large-scale networks.

Tools & Technologies

- **Simulator:** Custom Python Simulator
- **Algorithms:** Centrality Measures (CMBA), Reinforcement Learning (Q-Learning / Deep RL)
- **Languages:** Python
- **Evaluation Metrics:** Cache hit ratio, server hit reduction, hop reduction, latency, and throughput.

Conclusion

Our project aims to:

- Develop a scalable and intelligent topology optimization framework.
- Extend the centrality-based selection mechanism using a Large Action Model (LAM).
- Ultimately, create a more efficient Content-Centric Network for modern data demands.

References

Kumari Nidhi Lal, Anoj Kumar (2017).
A Centrality-measures based Caching Scheme for Content-centric Networking (CCN).
Multimedia Tools and Applications.

AI-Driven Dynamic Cache Policy Selection in Information-Centric Networking (FYP Thesis 2024-25).

Thank You

Questions?